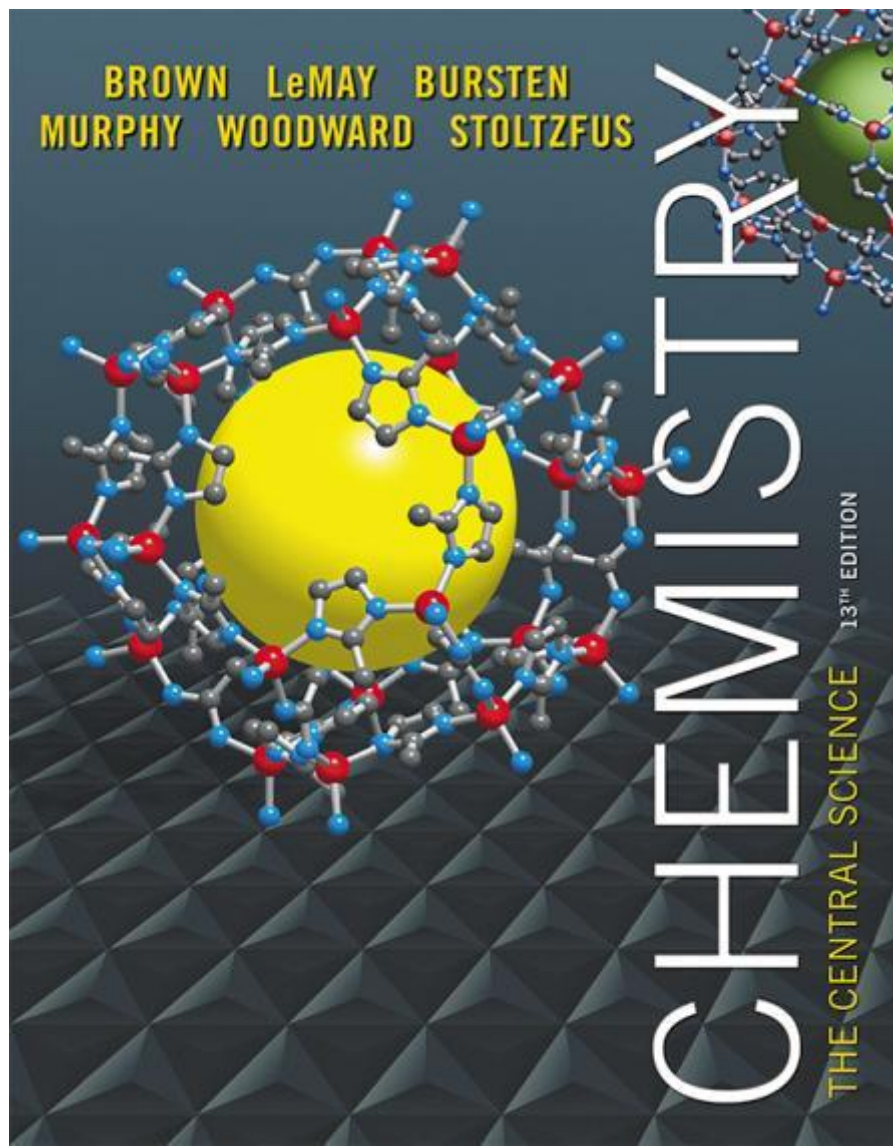


Chapter 1. Basic Concepts of Chemical Bonding

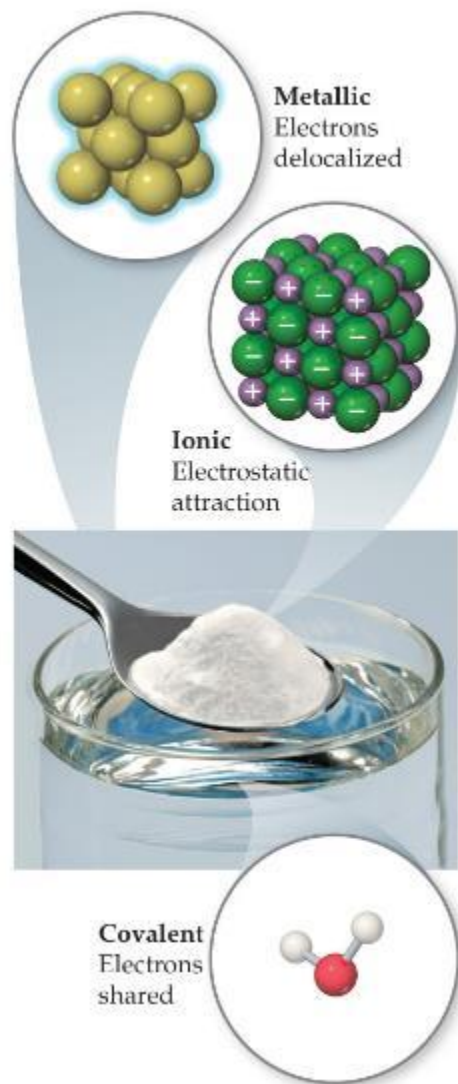


Reference:

Chapter 8

Basic Concepts of Chemical Bonding

Chemical Bonds

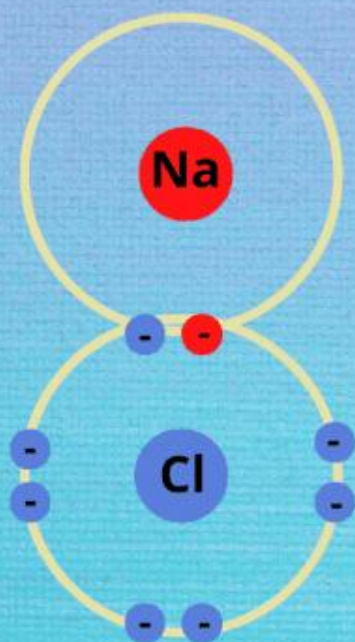


- Three basic types of bonds
 - Ionic 离子键
 - Electrostatic attraction between ions.
 - Covalent 共价键
 - Sharing of electrons.
 - Metallic 金属键
 - Metal atoms bonded to several other atoms.

Types of Chemical Bonds

Ionic Bond

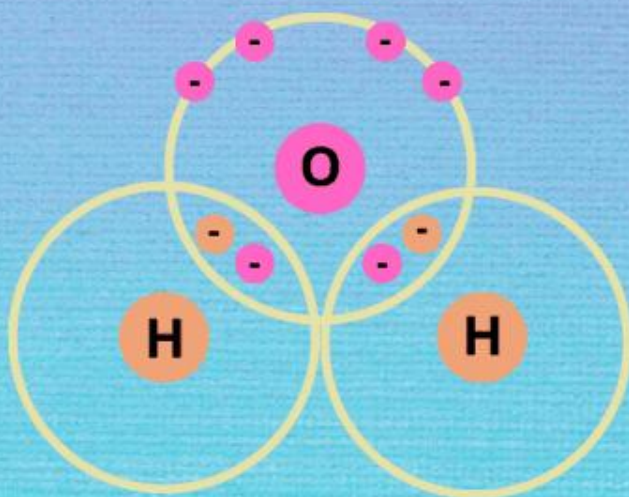
Metal atom donates electron to nonmetal atom.



NaCl

Covalent Bond

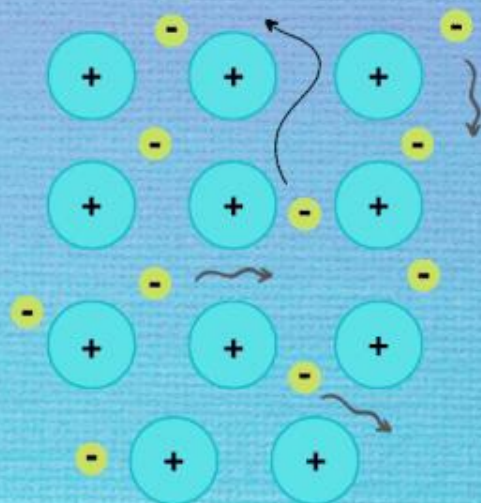
Two nonmetal atoms share electrons.



H₂O

Metallic Bond

Electrons move freely between metal atoms.



Fe

Lewis Symbols 路易斯符号

TABLE 8.1 Lewis Symbols

Element	Electron Configuration	Lewis Symbol	Element	Electron Configuration	Lewis Symbol
Li	[He]2s ¹	Li·	Na	[Ne]3s ¹	Na·
Be	[He]2s ²	·Be·	Mg	[Ne]3s ²	·Mg·
B	[He]2s ² 2p ¹	·B·	Al	[Ne]3s ² 3p ¹	·Al·
C	[He]2s ² 2p ²	·C·	Si	[Ne]3s ² 3p ²	·Si·
N	[He]2s ² 2p ³	·N·	P	[Ne]3s ² 3p ³	·P·
O	[He]2s ² 2p ⁴	:O:	S	[Ne]3s ² 3p ⁴	:S:
F	[He]2s ² 2p ⁵	·F:	Cl	[Ne]3s ² 3p ⁵	·Cl:
Ne	[He]2s ² 2p ⁶	:Ne:	Ar	[Ne]3s ² 3p ⁶	:Ar:

表示

- G. N. Lewis developed a method to denote potential bonding electrons by using one dot for every valence electron around the element symbol. 价电子
- When forming compounds, atoms tend to gain, lose, or share electrons until they are surrounded by **eight valence electrons (the octet rule)**. 化合物

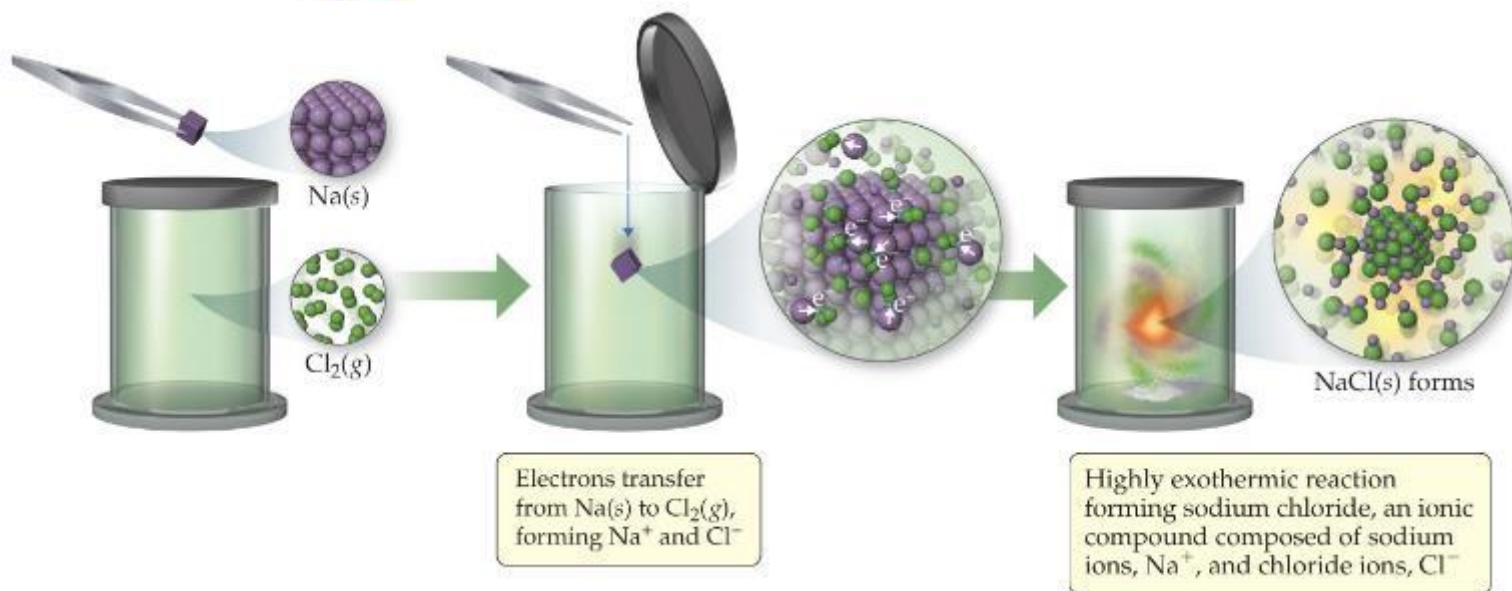
Ionic Formation

离子形成

- Atoms tend to **lose** (metals) or **gain** (nonmetals) electrons to make them isoelectronic to the noble gases.

同电子的

惰性气体



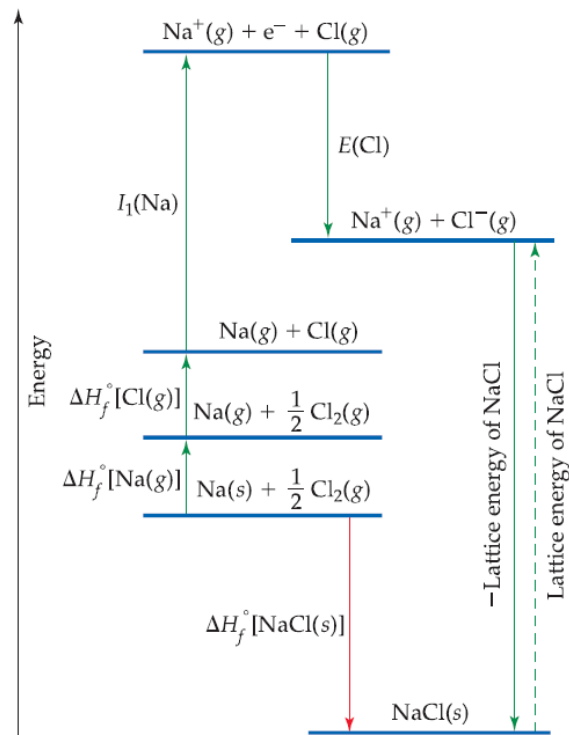
Energetics of Ionic Bonding— Born-Haber Cycle

伯恩-哈伯循环：用于分析离子化合物生成时的能量变化；测晶格能



▲ **Figure 8.3 The crystal structure of sodium chloride.** In this three-dimensional array of ions, each Na^+ ion is surrounded by six Cl^- ions, and each Cl^- ion is surrounded by six Na^+ ions.

- Many factors affect the energy of ionic bonding.
- Start with the metal and nonmetal elements: Na(s) and $\text{Cl}_2(\text{g})$.
- Make gaseous atoms: Na(g) and Cl(g) .
- Make ions: $\text{Na}^+(\text{g})$ and $\text{Cl}^-(\text{g})$.
- Combine the ions: NaCl(s) .



Energetics of Ionic Bonding

- We already discussed making ions (ionization ^{电离能} energy and electron affinity ^{电子亲和能：一个气态中性原子获得一个电子时所发生的能量变化}). ^{非金属（特别是卤素）有较大的电子亲和能}
- It takes energy to convert ^{转变} the elements to atoms. (endothermic) ^{吸热的}
- It takes energy to create a cation ^{阳离子} (endothermic).
- Energy is released by making the anion ^{阴离子} (exothermic). ^{放热的}
- The formation ^{形成} of the solid releases a *huge* amount of energy (exothermic).
- This makes the formation of salts from the elements *exothermic*.

Lattice Energy

晶格能

气态正、负离子结合形成 1 mol 固态离子化合物晶体 时所释放的能量

放热的

- That huge, exothermic transition is the reverse of the **lattice energy**,
 - *the energy required to completely separate a mole of a solid ionic compound into its gaseous ions.*
- The energy associated with electrostatic interactions is governed by Coulomb's law:

相反的

摩尔

气态的

约束

$$E_{\text{el}} = \kappa \frac{Q_1 Q_2}{d}$$

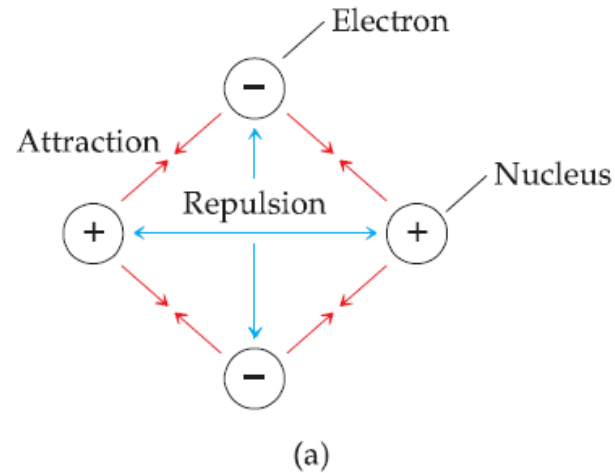
Lattice Energy

- Lattice energy increases with:
 - increasing charge on the ions
 - decreasing size of ions

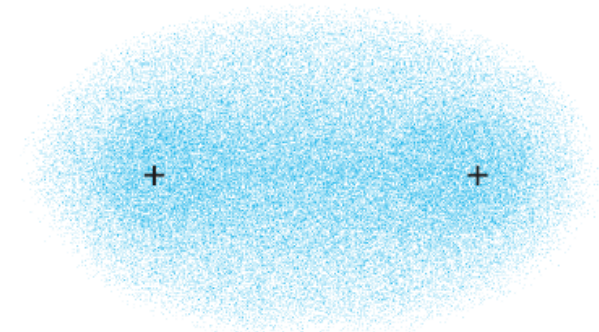
Table 8.2 Lattice Energies for Some Ionic Compounds

Compound	Lattice Energy (kJ/mol)	Compound	Lattice Energy (kJ/mol)
LiF	1030	MgCl ₂	2326
LiCl	834	SrCl ₂	2127
LiI	730		
NaF	910	MgO	3795
NaCl	788	CaO	3414
NaBr	732	SrO	3217
NaI	682		
KF	808	ScN	7547
KCl	701		
KBr	671		
CsCl	657		
CsI	600		

Covalent Bonding



(a)



(b)

- In covalent bonds, atoms share electrons.
- There are several electrostatic interactions in these bonds:
 - 引力 attractions between electrons and nuclei,
 - 排斥力 repulsions between electrons, and
 - repulsions between nuclei.
- For a bond to form, the attractions must be greater than the repulsions.

Lewis Structures

路易斯结构式

- Sharing electrons to make covalent bonds can be demonstrated using Lewis structures.
- We start by trying to give each atom the same number of electrons as the nearest noble gas by sharing electrons.
- The simplest examples are for hydrogen, H_2 , and chlorine, Cl_2 , shown below.



Electrons on Lewis Structures

- 未共用电子对 Lone pairs: electrons located on *only one* atom in a Lewis structure
- 键合电子对 Bonding pairs: shared electrons in a Lewis structure; they can be represented by two dots *or* one line

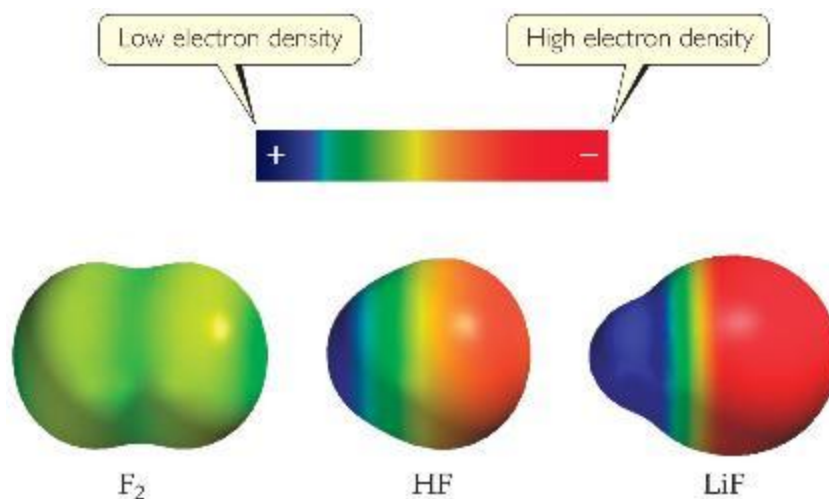
Multiple Bonds

- Some atoms share only one pair of electrons. These bonds are called **single bonds**.
- Sometimes, two pairs need to be shared. These are called **double bonds**.
- There are even cases where three bonds are shared between two atoms. These are called **triple bonds**.



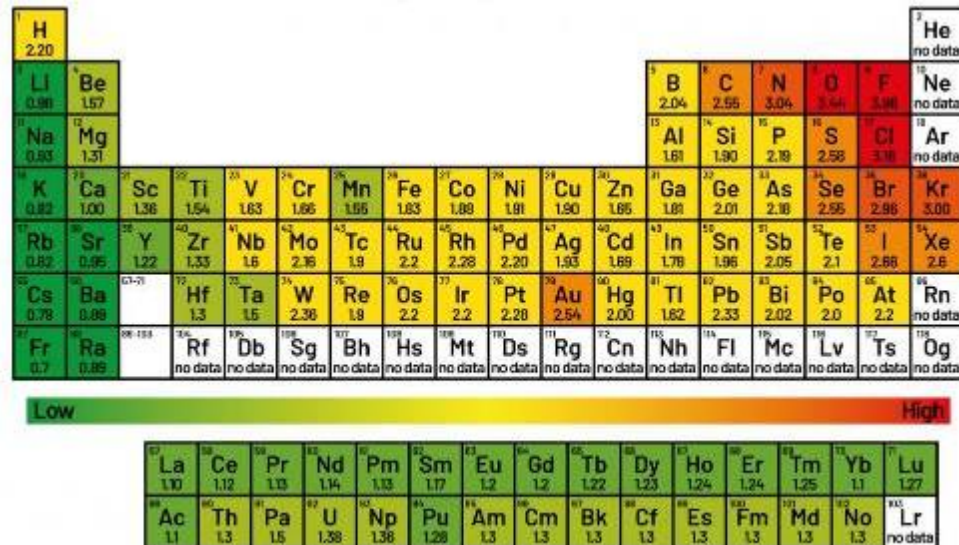
Polar Covalent Bonds 极性共价键

- The electrons in a covalent bond are not always shared equally.
- Fluorine pulls harder on the electrons it shares with hydrogen than hydrogen does.
- Therefore, the fluorine end of the molecule has more electron density than the hydrogen end. 分子



Electronegativity 电负性

- Electronegativity is the ability of an atom in a molecule to attract electrons to itself.
- On the periodic table, electronegativity generally increases as you go
 - from left to right across a period.
 - from the bottom to the top of a group.

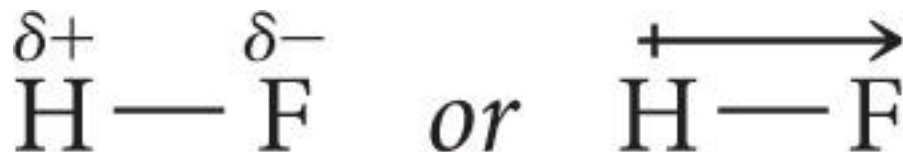


Electronegativity

- Electronegativity can tell us a lot about how different elements will bond to each other and which **type of bond** it will be.
- If the electronegativity difference between the two elements involved in bonding is **less than 0.4** then the **bond will be nonpolar covalent.** 非极性共价键
- If the difference is between **0.4 and 1.7** then the bond is **considered polar covalent.** 极性共价键
- And finally, if the difference is **greater than 1.7** then the **bond will be ionic.** 离子键

Electronegativity and Polar Covalent Bonds

- When two atoms share electrons unequally, a **polar covalent bond** results.
- Electrons tend to spend more time around the more electronegative atom. The result is a partial negative charge (部分负电荷) (*not* a complete transfer of charge). It is represented by δ^- .
- The other atom is “more positive,” or δ^+ .

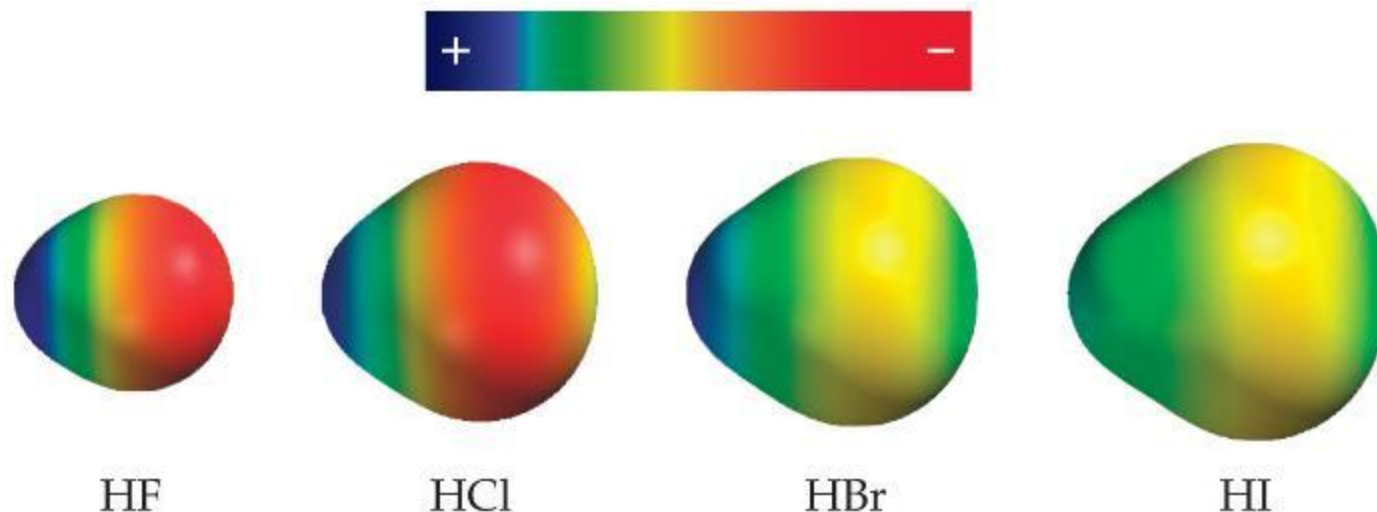


Polar Covalent Bonds

Table 8.3 Bond Lengths, Electronegativity Differences, and Dipole Moments of the Hydrogen Halides

Compound	Bond Length (Å)	Electronegativity Difference	Dipole Moment (D)
HF	0.92	1.9	1.82
HCl	1.27	0.9	1.08
HBr	1.41	0.7	0.82
HI	1.61	0.4	0.44

The greater the difference in electronegativity, the more polar is the bond.



Dipoles

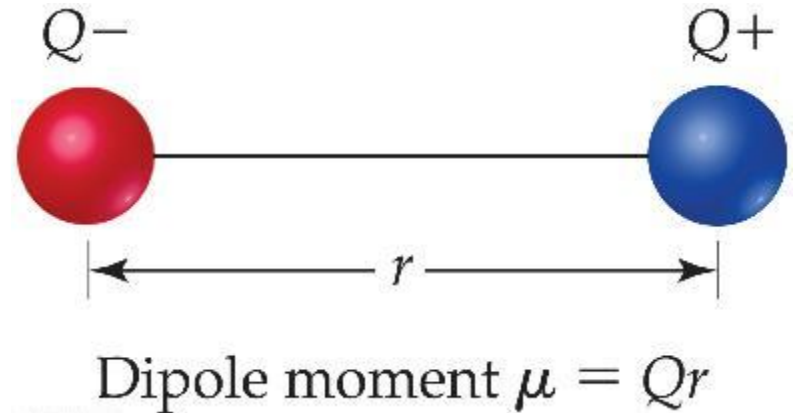
二极体\偶极

当一个分子或化学键中，电荷分布不均匀时，会形成部分正电荷 (δ^+) 和部分负电荷 (δ^-)，这对电荷就叫偶极

- When two equal, but opposite, charges are separated by a distance, a **dipole** forms.
- A **dipole moment**, μ ,
偶极距
produced by two equal but opposite charges separated by a distance, r , is calculated:
- It is measured in debyes (D).

$$\mu = Qr$$

单位



Is a Compound Ionic or Covalent?

- Simplest approach: Metal + nonmetal is ionic; nonmetal + nonmetal is covalent. 离子键 共价键
- There are many exceptions: It doesn't take into account oxidation number of a metal (higher oxidation numbers can give covalent bonding). 例外 氧化价, 化合价
- Electronegativity difference can be used; the table still doesn't take into account oxidation number.
- Properties of compounds are often best: Lower melting points mean covalent bonding, for example.

Writing Lewis Structures (Covalent Molecules)

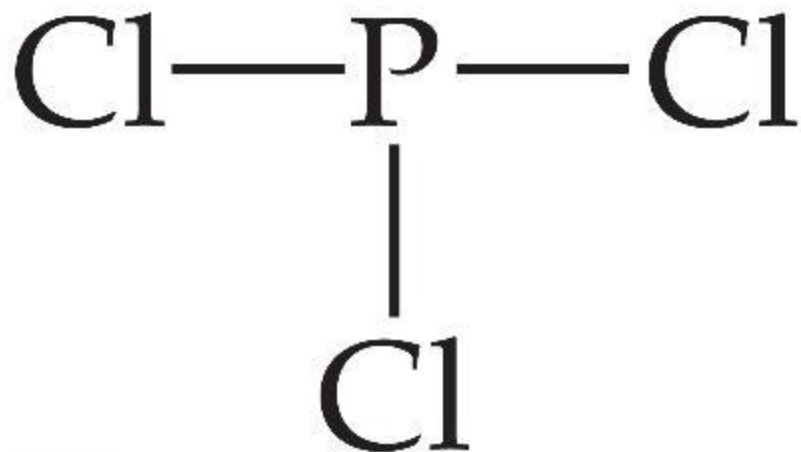


1. Sum the valence electrons from all atoms, taking into account overall charge.
 - If it is an anion, add one electron for each negative charge.
 - If it is a cation, subtract one electron for each positive charge.

Keep track of the electrons:

$$5 + 3(7) = 26$$

Writing Lewis Structures

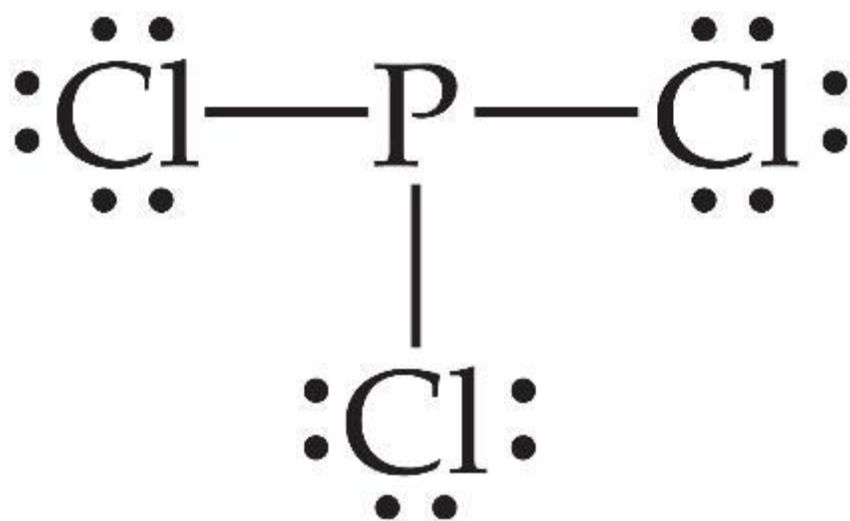


2. Write the symbols for the atoms, show which atoms are attached to which, and connect them with a single bond (a line representing two electrons).

Keep track of the electrons:

$$26 - 6 = 20$$

Writing Lewis Structures

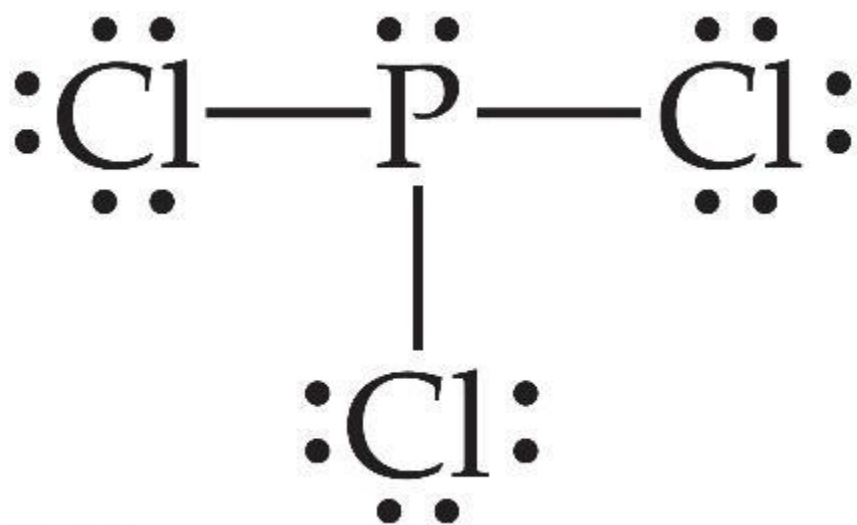


3. Complete the octets around all atoms bonded to the central atom.

Keep track of the electrons:

$$26 - 6 = 20; 20 - 18 = 2$$

Writing Lewis Structures



4. Place any leftover electrons on the central atom.

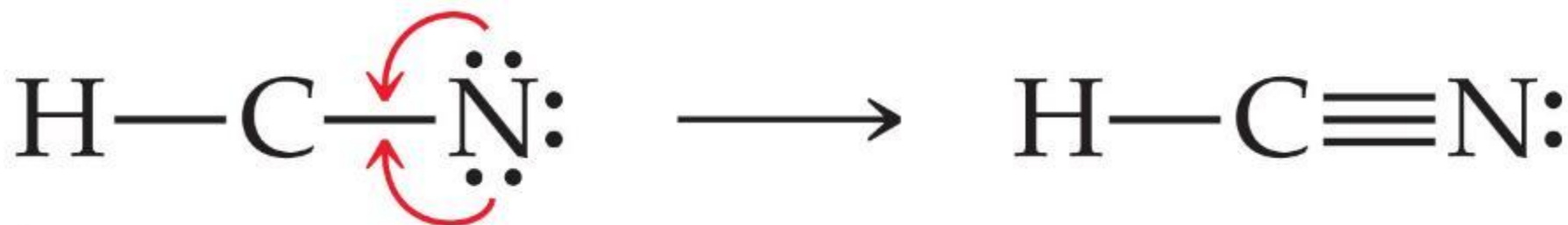
Keep track of the electrons:

$$26 - 6 = 20; 20 - 18 = 2; 2 - 2 = 0$$

Writing Lewis Structures

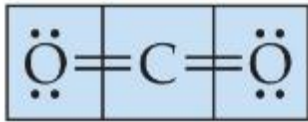
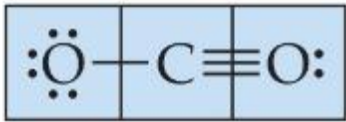
5. *If there are not enough electrons to give the central atom an octet, try multiple bonds.*

满八电子



Writing Lewis Structures

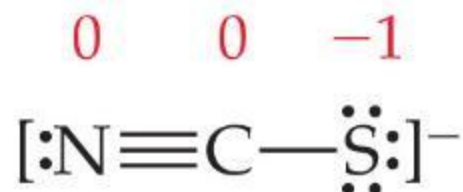
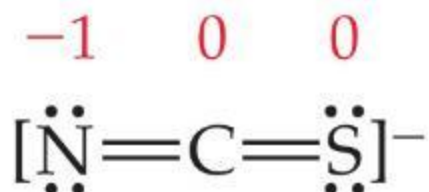
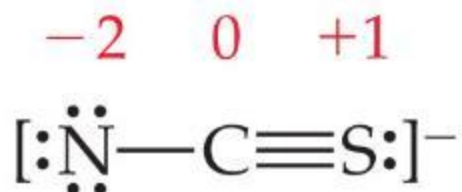
- Then assign formal charges. 形式电荷 (Formal Charge, FC) 是一种假想电荷，用于估算原子在分子或离子中理论上的电荷分布。
- Formal charge** 形式电荷 is the charge an atom would have if all of the electrons in a covalent bond were shared equally. 它假设所有键中电子平均分配给参与键的原子，用于判断 Lewis 结构的合理性。
- Formal charge = valence electrons 价电子 $-\frac{1}{2}$ (bonding electrons) 形成化学键的电子 $-$ all nonbonding electrons

						
Valence electrons:	6	4	6	6	4	6
-(Electrons assigned to atom):	6	4	6	7	4	5
Formal charge:	0	0	0	-1	0	+1

原子原来电子数-外电子数，越接近0，越稳定，则为路易斯结构

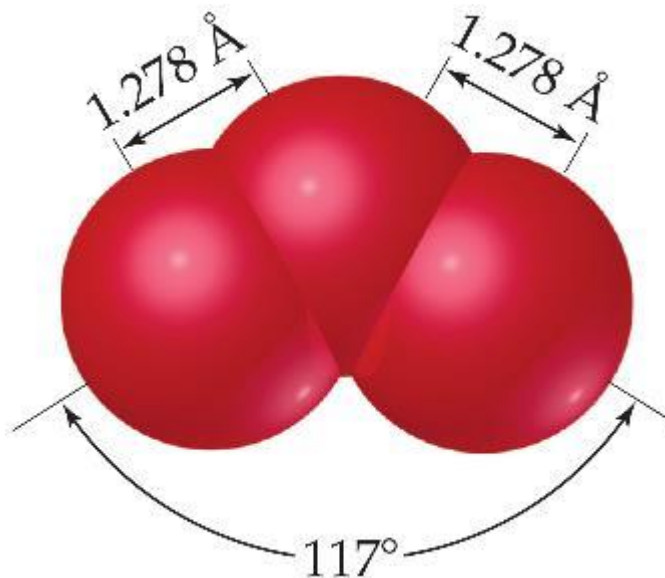
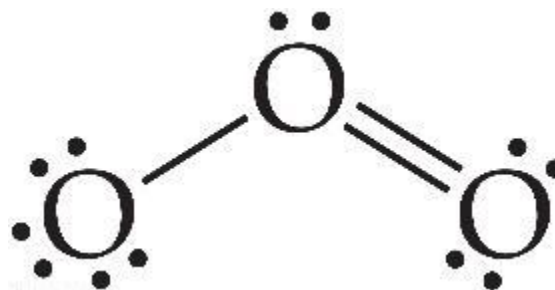
Writing Lewis Structures

- The dominant Lewis structure
 - is the one in which atoms have formal charges closest to zero.
 - puts a negative formal charge on the most electronegative atom.



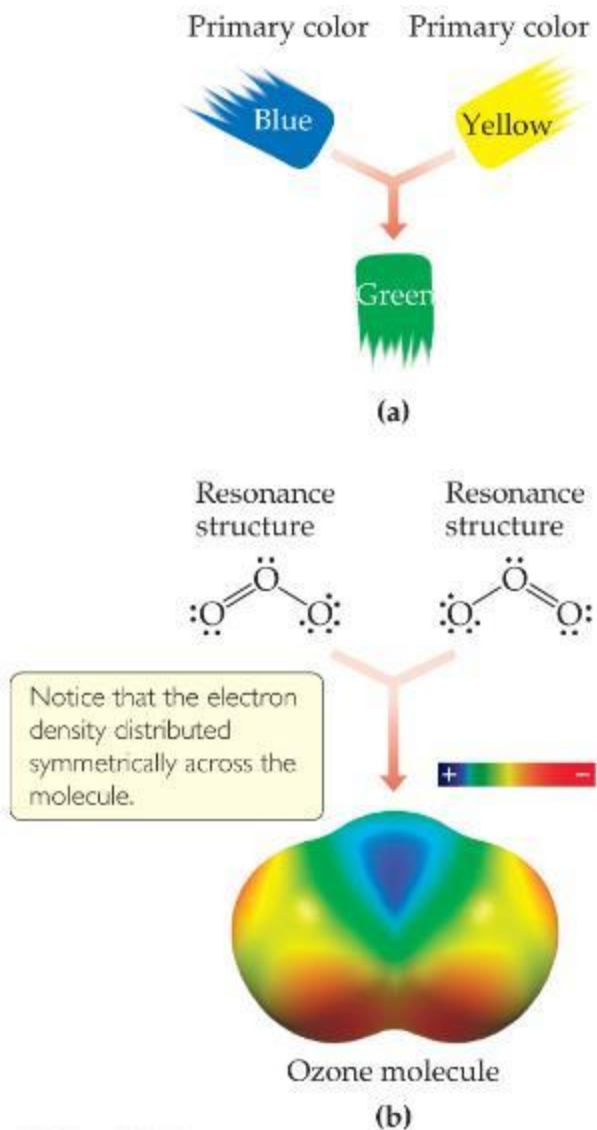
The Best Lewis Structure?

- Following our rules, this is the Lewis structure we would draw for ozone, O_3 .
- However, it doesn't agree with what is observed in nature: Both O to O connections are the same.

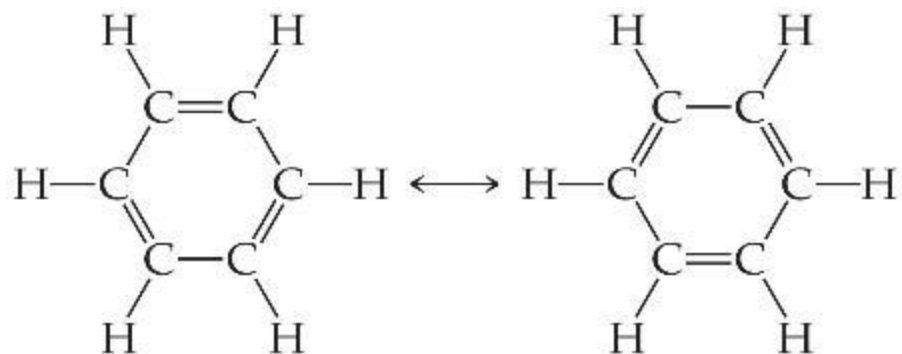


Resonance 共振

- One Lewis structure cannot accurately depict a molecule like ozone.
- We use multiple structures, **resonance structures**, to describe the molecule.



Resonance



- The organic compound benzene, C_6H_6 , has two resonance structures.
- It is commonly depicted as a hexagon with a circle inside to signify the delocalized electrons in the ring. 表示

Localized electrons are specifically on one atom or shared between two atoms; *Delocalized* electrons are shared by multiple atoms.

Exceptions to the Octet Rule

八隅体规则

- There are three types of ions or molecules that do not follow the octet rule:
 - ions or molecules with an odd number of electrons,
 - ions or molecules with less than an octet,
 - ions or molecules with more than eight valence electrons (an expanded octet).

奇数

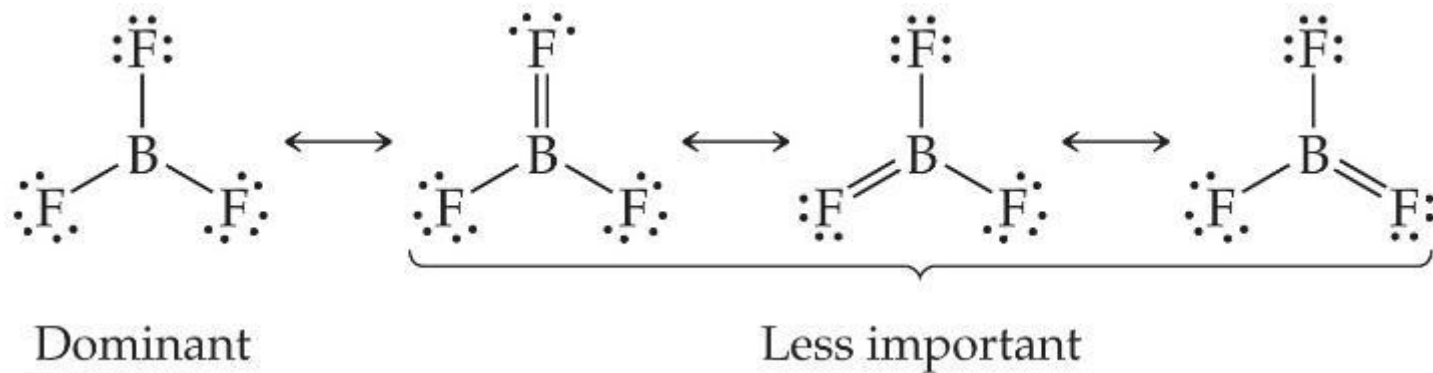
Odd Number of Electrons

Though relatively rare and usually quite unstable and reactive, there are ions and molecules with an odd number of electrons.



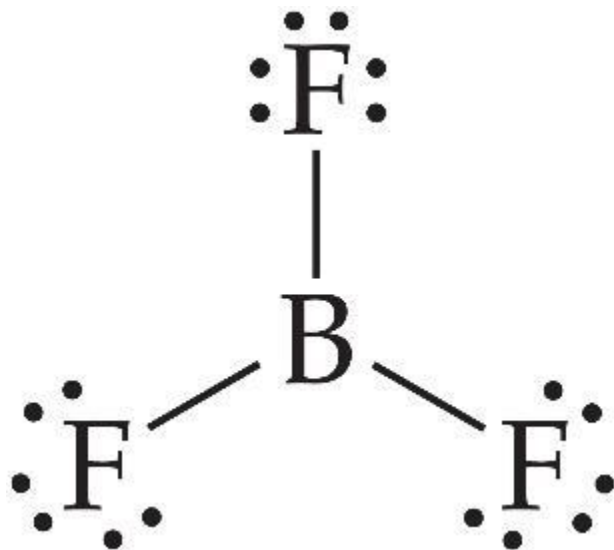
Fewer Than Eight Electrons

- Elements in the second period *before* carbon can make stable compounds with fewer than eight electrons.
- Consider BF_3 :
 - Giving boron a filled octet places a *negative* charge on the boron and a *positive* charge on fluorine.
 - This would not be an accurate picture of the distribution of electrons in BF_3 .



Fewer Than Eight Electrons

The lesson is: If filling the octet of the central atom results in a negative charge on the central atom and a positive charge on the more electronegative outer atom, don't fill the octet of the central atom.



BF₃为何不满八电子

Sep 18, 2025 • Content generated by AI. Might not be accurate.

为什么bf3可以形成稳定的分子但其中的原子不满八电子

BF₃ 能稳定存在但中心原子不满八电子，是因为**硼的特殊电子结构和共振效应**共同作用的结果：

1. 硼的电子结构特点

- 硼原子只有3个价电子
- 形成BF₃时，硼与3个氟原子形成3个共价键
- 价层电子总数只有6个，未达到八电子稳定结构

2. 氟的强电负性作用

- 氟原子强烈吸引电子，使B-F键具有明显极性
- 这种极性键产生的静电吸引力部分补偿了八电子缺失的不稳定性

3. 共振稳定效应

- BF₃存在共振结构，负电荷可在三个氟原子间离域
- 共振使分子整体能量降低，提高了稳定性

4. 空轨道特性

- 硼原子有一个空的2p轨道
- 这使BF₃成为路易斯酸，能接受电子对形成配位化合物
- 例如与NH₃形成F₃B:NH₃配合物，使硼达到八电子结构

BF₃虽然不满足八电子规则，但通过极性共价键和共振效应获得了足够的稳定性，同时其空轨道特性也解释了为什么它容易参与化学反应。

想了解更多关于价层电子对互斥理论(VSEPR)如何预测BF₃的分子构型吗？



Doubao

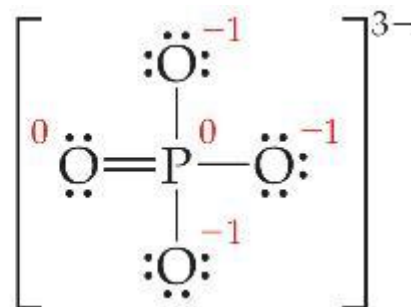
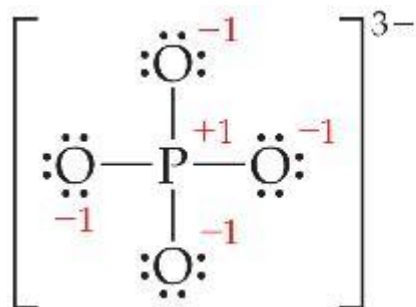
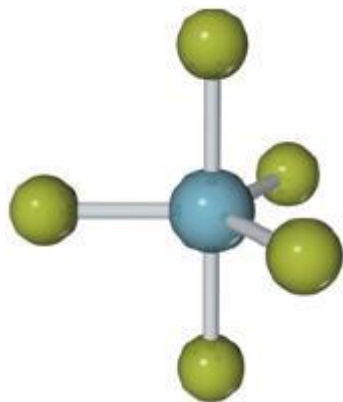
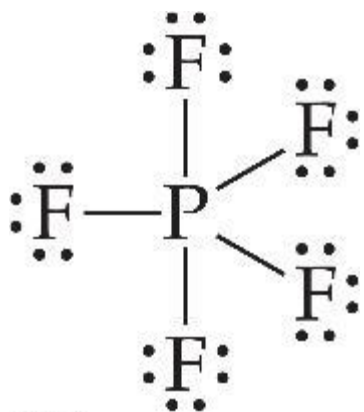
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More Than Eight Electrons

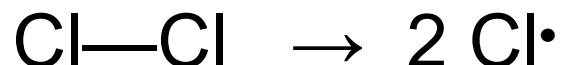
- When an element is in period 3 or below in the periodic table (e.g., periods 3, 4, 5, etc.), it can use *d*-orbitals to make *more* than four bonds.
- Examples: PF_5 and phosphate below

(Note: Phosphate will actually have four resonance structures with five bonds on the P atom!)



Covalent Bond Strength

- Most simply, the strength of a bond is measured by determining how much energy is required to break the bond.
- This is called the **bond enthalpy**. 键焓
- The bond enthalpy for a Cl—Cl bond, $D(\text{Cl—Cl})$, is measured to be 242 kJ/mol.
- We write out reactions for breaking *one mole* of those bonds:



Average Bond Enthalpies

- Average bond enthalpies are positive, because bond breaking is an endothermic process.
- Note that these are *averages* over many different compounds; not every bond in nature for a pair of atoms has exactly the same bond energy.

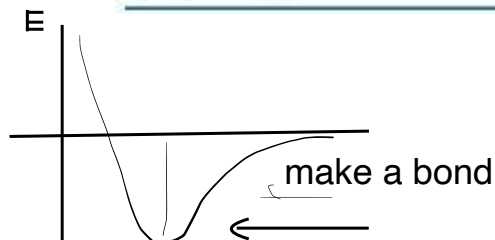
Table 8.4 Average Bond Enthalpies (kJ/mol)

Single Bonds

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						
Si—Cl	464						

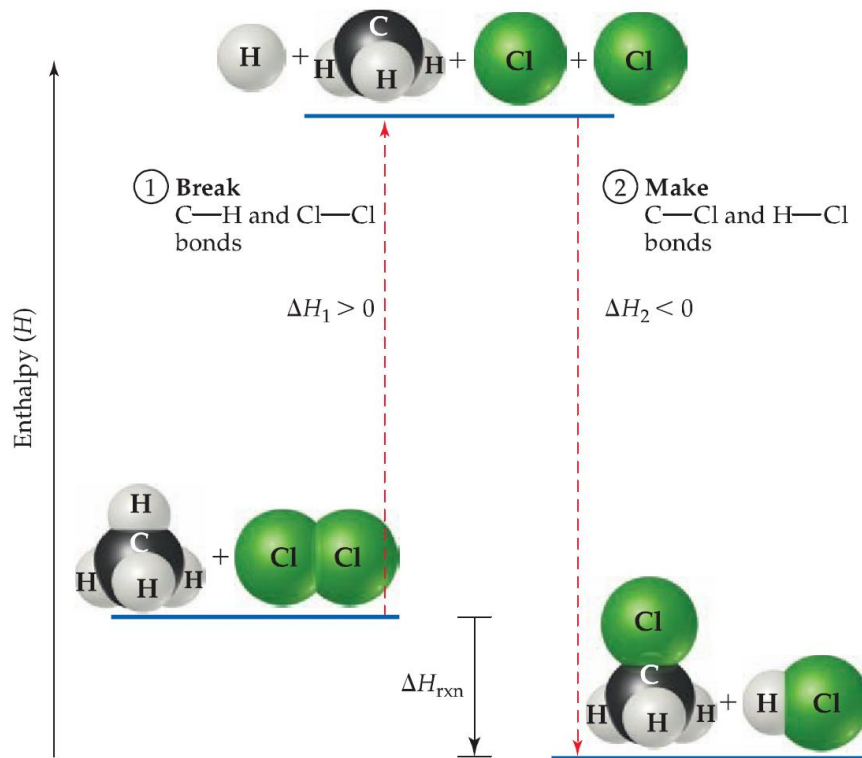
Multiple Bonds

C=C	614	N=N	418	O=O	495
C≡C	839	N≡N	941		
C=N	615	N=O	607	S=O	523
C≡N	891			S=S	418
C=O	799				
C≡O	1072				



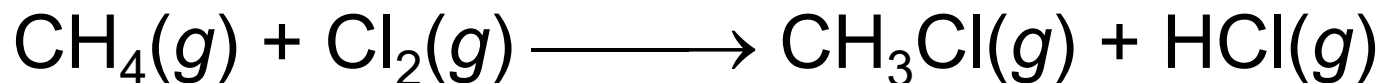
Using Bond Enthalpies to Estimate Enthalpy of Reaction

- One way to *estimate* ΔH for a reaction is to use the bond enthalpies of bonds broken and the new bonds formed.
- Energy is added to break bonds and released when making bonds.
- In other words, $\Delta H_{rxn} = \Sigma(\text{bond enthalpies of all bonds broken}) - \Sigma(\text{bond enthalpies of all bonds formed})$.



Example

From the figure on the last slide



- In this example, one C—H bond and one Cl—Cl bond are broken; one C—Cl and one H—Cl bond are formed.

Answer

$$\begin{aligned}\Delta H &= [D(\text{C—H}) + D(\text{Cl—Cl})] - [D(\text{C—Cl}) + D(\text{H—Cl})] \\ &= [(413 \text{ kJ}) + (242 \text{ kJ})] - [(328 \text{ kJ}) + (431 \text{ kJ})] \\ &= (655 \text{ kJ}) - (759 \text{ kJ}) \\ &= -104 \text{ kJ}\end{aligned}$$

Bond Enthalpy and Bond Length

- We can also measure an average bond length for different bond types.
- As the number of bonds between two atoms increases, the bond length decreases.

Table 8.5 Average Bond Lengths for Some Single, Double, and Triple Bonds

Bond	Bond Length (Å)	Bond	Bond Length (Å)
C—C	1.54	N—N	1.47
C=C	1.34	N=N	1.24
C≡C	1.20	N≡N	1.10
C—N	1.43	N—O	1.36
C=N	1.38	N=O	1.22
C≡N	1.16		
		O—O	1.48
C—O	1.43	O=O	1.21
C=O	1.23		
C≡O	1.13		